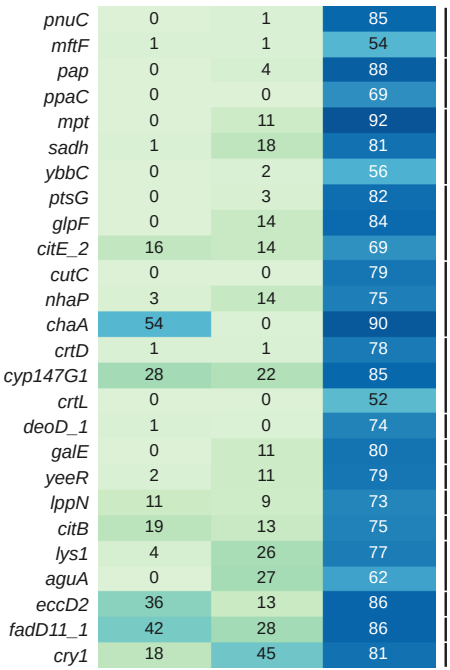
Candidate FSM-associated gene families by functional category



	Coenzyme metab	nicotinamide mononucleotide transporter (KO K03811)
		family-2 glycosyltransferase, mycofactocin-associated (Pfam Glycos_transf_2)
	Energy production	PPK2-family polyphosphate kinase (Pfam PPK2; symbol 'pap' is eggNOG's)
		Mn-dependent inorganic pyrophosphatase (KO K15986)
	Other/unknown	fasciclin-domain protein (Pfam Fasciclin)
		zinc-dependent alcohol dehydrogenase (Pfam ADH_N + ADH_zinc_N)
		DUF1343 domain protein
	Carbohydrate metab	glucose-family PTS permease, single-chain EIICBA (KO K02757)
		MIP-family glycerol/water channel (KO K02440, glycerol uptake facilitator)
		HpcH/HpaI-family aldolase, citrate lyase beta-subunit type (KO K01644)
	Ion transport	copper homeostasis protein CutC (KO K06201)
		Na+/H+ exchanger family (Pfam Na_H_Exchanger)
		Ca2+:H+ (Na+/Ca2+) antiporter (KO K07300)
	Secondary metabolites	carotenoid desaturase, CrtI/CrtD family (FAD-dependent; KO K09845)
		cytochrome P450 (Pfam p450)
		lycopene beta-cyclase (Pfam Lycopene_cycl; KO K06443)
	Nucleotide metab	nucleoside phosphorylase, PNP/UDP family (KO K00757, uridine phosphorylase)
	Cell wall/envelope	UDP-glucose 4-epimerase (KO K01784)
	PTM/chaperones	membrane methyltransferase, PEMT/ICMT-like (Pfam PEMT; KO K21310 mddA)
	No COG assignment	uncharacterised protein (symbol lppN from eggNOG; no domain, KO or COG hit)
	Signalling	CitB-family two-component response regulator (KO K02475)
	Amino-acid metab	saccharopine dehydrogenase (KO K00290; Pfam Sacchrp_dh)
		agmatine deiminase (KO K10536)
	Secretion	T7SS ubiquitin-like protein (Pfam YukD)
	Lipid metab	long-chain-fatty-acid--CoA ligase (KO K01897; Pfam AMP-binding)
	Replication/repair	photolyase/cryptochrome family (Pfam DPRP + FAD_binding_7)

A selected subset, not a validated functional signature. Of the 99 candidate families in Table S13, 59 map to a discriminating MMseqs2 cluster (families file); 33 of these 59 have an FSM prevalence at least 30 points above the higher of RGM and SGM, and the three largest gaps per COG category are drawn (26 families). Prevalence is over 134 FSM, 389 RGM and 299 SGM genomes (Table S13 uses 96 SPIRE FSM MAGs and pools RGM + SGM). Labels are short descriptions checked against Pfam / KEGG-KO evidence of the cluster representative.